

Gaming Habits and Academic Performance: Dimensional Reduction and Clustering

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Problem 1: Dataset Selection

This project uses the Gaming Habits and Academic Performance dataset from Kaggle, which contains 8,000 college students and 14 columns. It was originally selected for our INFO 432 group project and is reused here with approval. The columns cover gaming behavior (daily gaming hours, genre, addiction score, device usage), lifestyle factors (study hours, sleep hours, attendance, social activity), physiological and psychological measures (reaction time, stress level), and academic outcome (grades). Video games are frequently blamed for poor academic performance, but the relationship is probably more nuanced than the simple claim that more gaming leads to worse grades. This dataset is a good fit for testing that directly, and for this assignment specifically, it gives a mix of behavioral and academic variables that plausibly split into distinct underlying groups, which is exactly what PCA, CCA, and clustering are built to find.

Going into the dimensional reduction step, the gaming-related variables (gaming hours, device usage, reaction time, addiction score) were expected to load heavily onto a shared component, since they likely reflect one underlying screen engagement factor. Study-related and wellbeing variables (study hours, sleep, attendance) were expected to form a separate component. Using CCA to relate a gaming and behavior variable set against a lifestyle and academic variable set, a moderate canonical correlation was expected, since gaming and lifestyle habits are related but should not fully overlap. For the clustering step, unsupervised clustering was expected to reveal distinct student profiles, such as a high gaming and low study cluster versus a low gaming and high study cluster, without ever being told grades. Clusters formed on PCA-reduced features were also expected to be more distinct than clusters formed on the raw variables, since PCA should strip out noise before clustering runs.

The exploratory analysis script loads the dataset, checks structure and missing values, of which there are none, and summarizes distributions for every numeric variable. A correlation matrix against all thirteen numeric features confirmed the pattern the group project had already found: gaming hours and addiction score correlate negatively with grades, study hours correlates positively, and device usage and reaction time move closely with gaming hours. This correlation structure directly motivated the groupings used in the dimensional reduction step that follows.

Problem 2: Dimensional Reduction

Two methods were fit for this problem: Principal Component Analysis on all thirteen numeric features, and Canonical Correlation Analysis relating a gaming and behavior variable set against a lifestyle and academic variable set. All variables were standardized to a mean of zero and a standard deviation of one before fitting, since PCA is sensitive to scale and the raw variables span very different ranges, with reaction time in the hundreds of milliseconds and gaming hours ranging from zero to eight.

The first principal component explains 39.7 percent of the total variance on its own, and is driven almost entirely by gaming hours, addiction score, device usage, and reaction time, exactly the screen engagement grouping hypothesized above. The second component explains an additional 15.7 percent and is driven by study hours, grades, and sleep hours, an academic effort axis that also matches the original hypothesis. Together, these first two components capture 55.4 percent of the total variance in the dataset using two numbers instead of thirteen, which is a meaningful reduction. The remaining components each explain roughly nine to ten percent and do not correspond to an obviously interpretable grouping, so the best fitting solution here is the first two components.

For the canonical correlation analysis, the gaming and behavior set, made up of gaming hours, device usage, reaction time, and addiction score, was related against the lifestyle and academic set, made up of study hours, sleep hours, attendance, and social activity. The first and strongest canonical correlation came out to only 0.035, which is essentially no linear relationship between the two sets treated as whole groups. This directly refutes the moderate correlation that was hypothesized above. The result is not a contradiction of the correlation matrix, however. Individual pairs within these sets, such as gaming hours and grades, are still strongly related on their own. What the canonical correlation shows is that the two variable sets do not move together as coordinated groups, meaning the relationship between gaming and academics in this dataset is carried by a few specific variable pairs rather than by a broad multivariate relationship between the two domains.

Problem 3: Unsupervised Clustering

Two clustering methods were fit for this problem, K-means and hierarchical clustering using Ward's method, each applied to both the raw standardized features and the four principal components from the previous step, for four combinations in total. A value of k equal to four was chosen using the elbow method on the raw scaled data. Cluster quality was compared using the average silhouette score, which ranges roughly from negative one to positive one,

with higher values indicating better separated clusters.

K-means on the raw scaled features produced an average silhouette score of 0.147, while K-means on the PCA-reduced features scored 0.220, the best result overall. Hierarchical clustering scored 0.099 on the raw features and 0.177 on the PCA-reduced features. In both cases, the PCA-reduced features produced better separated clusters than the raw features, and K-means outperformed hierarchical clustering. More importantly, the four K-means clusters found on the raw features, with zero knowledge of any student's actual grade, split students into groups averaging 39.0, 60.7, 76.0, and 91.0 in grade. Clustering on behavior alone recovered something that closely resembles real academic performance tiers, which is a strong and easily explainable result, since each cluster can be described in one sentence, for example a high gaming, low study group averaging an F, without needing to reference the grade column at all.

Going into this assignment, the gaming and lifestyle variables were expected to behave like two cleanly separated groups, both in the principal component loadings and in the canonical correlation. Half of that expectation held up, and half of it did not, in a way that actually clarified how this dataset works rather than just confirming what was already assumed.

The principal component results matched the original hypothesis almost exactly. The first component turned out to be a gaming and screen engagement axis, with gaming hours, addiction score, device usage, and reaction time all loading together, and the second component turned out to be an academic effort axis, with study hours, grades, and sleep. The variables expected to group together did so without needing to force the interpretation after the fact, and the two components together captured over half of the dataset's total variance. That was a genuinely satisfying result, since it meant the intuitive story about this dataset, gaming habits on one side and academic habits on the other, is actually reflected in the underlying structure of the data and not just in surface level correlations.

The canonical correlation result is where the hypothesis was wrong, and where the most was learned. A moderate canonical correlation was expected between the gaming set and the lifestyle set, on the reasoning that if gaming hurts grades, the two domains should move together to some degree as coordinated groups. Instead, the first canonical correlation came out to 0.035, functionally zero. The first instinct was that something had gone wrong in the model, since the correlation matrix and the group project's regression model had already shown that gaming hours and grades are meaningfully related. Those two results are not actually in

conflict, though. Canonical correlation tests whether two whole sets of variables move together as linear combinations of each other, not whether any individual pair within them is related. What this shows is that the relationship between gaming and academic outcomes in this dataset is carried by specific variables, not by gaming behavior and lifestyle behavior treated as broad, interchangeable categories. That is a more precise and, honestly, more useful finding than the vaguer version that was hypothesized going in.

The clustering results were the most rewarding part of this assignment. Fitting K-means and hierarchical clustering on both the raw features and the PCA-reduced features gave a direct, numeric answer to which feature set clusters better: the PCA-reduced features won for both methods, and K-means beat hierarchical clustering both times. That confirmed the hypothesis about PCA-reduced clustering being cleaner, and it made intuitive sense once seen, since PCA had already stripped away redundant, correlated noise before clustering ever ran, leaving the clustering algorithm with less irrelevant structure to get confused by. What was surprising was how interpretable the resulting clusters turned out to be. K-means was fit with zero access to the grades column, purely on gaming and lifestyle behavior, and the four clusters it found still separated into average grades of 39, 61, 76, and 91. It was not expected that behavioral clustering alone would recover something this close to real performance tiers. It is a good reminder that even without supervision, if the underlying behavioral patterns are strong enough, they will show up in an outcome variable the model never even saw.

Overall, working with this dataset for a second time, after already having used it for the group project's regression and threshold analysis, provided a genuinely different angle on the same data. The regression model showed that gaming hurts grades and that the effect is worse for low-study students. The principal component and clustering analysis showed something complementary: that gaming behavior and academic behavior are each internally coherent groups of variables, that they do not move together as tightly as expected at the group level, and that behavioral clustering alone is enough to recover real performance differences without ever looking at a grade. Extending this further would mean testing whether a third or fourth canonical variate pair captures any meaningful signal at all, since they were smaller still at 0.014 and below, and probably do not, and trying a higher value of k for clustering to see whether a finer grained split, such as six or eight clusters, reveals subgroups within the current four, like a high gaming group that still manages to study enough to avoid the worst outcomes.

This dataset and its initial framing were developed collaboratively with the INFO 432 group project teammates, Jeffrey Cheung, Drew Jones, and Joel Thomas. The dimensional

reduction and clustering analysis in this submission is individual work.